

# Kunal Maheshwari

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## Summary

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**Applied AI / Backend Developer** with 4.6+ years of enterprise software experience and an MS in Computer Science, extending a Java/API foundation through **Python, FastAPI, PyTorch, Hugging Face Transformers, T5, and PEFT/LoRA** projects.

**Delivery scope:** Builds AI-backed application workflows from model fine-tuning and inference through FastAPI services, REST integration, React clients, evaluation, diagnostics, and containerized delivery.

**Engineering approach:** Separates application and model boundaries, manages per-user model adapters, measures output quality, and adds storage and observability suited to privacy-conscious AI workflows.

**Working style:** Learns new technologies through hands-on prototypes, tests assumptions against measurable behavior, documents reusable solutions, and collaborates across engineering, product-facing, QA, and data-dependent teams.

## Technical Skills

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**AI/ML & Generative AI:** Python, PyTorch, Hugging Face Transformers, T5, PEFT/LoRA, sequence-to-sequence models, fine-tuning and inference, local LLM concepts, ROUGE evaluation.

**AI Application Development:** FastAPI model serving, email summarization and classification, adapter lifecycle, MinIO model-artifact storage, Prometheus diagnostics; RAG/local-model assistant (in-progress project).

**APIs & Product Development:** REST APIs, Java 8/11, Spring Boot/Framework, JavaScript/TypeScript (projects), React, Vite, Tailwind CSS, XML/JSON.

**Data, Tools & Delivery:** PostgreSQL, SQL, Docker, Git, GitHub, GitHub Actions, Maven, Jenkins, CI/CD, JUnit, Postman, Agile/Scrum, technical documentation.

## Selected AI & Software Projects

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### Privacy-Preserving AI Email Client (Graduate Capstone)

- Built a privacy-oriented email summarization and classification product with a **Python/FastAPI model service and React/TypeScript client**, plus Docker, MinIO, and Prometheus.
- Implemented **T5-compatible fine-tuning and inference**, PEFT/LoRA adapter workflows, categorization, diagnostics, and per-user adapter loading with lifecycle controls.
- Evaluated the fine-tuned model at **ROUGE-1 F 53.04, ROUGE-2 F 44.90, and ROUGE-L F 45.90**, retaining measurable output-quality evidence.

### Local Programming Explanation Assistant (In Progress)

- Prototyping a **Python/React RAG workflow** for code and source explanation using local transformer hosting, modular Docker containers, and a separate database boundary.

### Interactive Portfolio System

[github.com/preferablehuman/portfolio](https://github.com/preferablehuman/portfolio)

- Built and deployed a React/TypeScript SPA with searchable skill evidence, responsive navigation, deep-link fallback, and automated GitHub Actions delivery.

## Work Experience

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NTT Data, Bengaluru, India

*Software Developer, Aug 2022 - Aug 2024*

- Developed Java/Spring and REST/SOAP healthcare integration services supporting **50K+ daily requests at 99.99% SLA availability**, strengthening API design and production reliability fundamentals.
- **Led a project-wide Log4j2 migration** from PoC through documented rollout, preserving production behavior and reviewing team implementation changes.
- Integrated correlation-ID monitoring for latency, success/failure, and volume telemetry; supported JUnit/API testing, CI/CD validation, and cross-system debugging.

**Capgemini**, Mumbai, India

*Senior Analyst, Jul 2020 - Aug 2022*

- Integrated an asynchronous **Python prediction service** with Java application APIs and callbacks, retaining deterministic Java validation before recommendations reached dealer users.
- Reduced VIN repeat-configuration API latency from **60+ seconds to 10–20 seconds** through HQL/DB2 query restructuring while preserving application business rules.
- Diagnosed defects across Python output, Java logs, and DB2 data; contributed testing, API validation, documentation, code reviews, and junior-developer mentoring.

**Capgemini**, Mumbai, India

*Analyst Trainee, Jan 2020 - Jun 2020*

- Built Java/Spring, Angular, and PostgreSQL modules for a digital-wallet application, including authentication, balances, transactions, and profile workflows.

## **Education**

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**Central Michigan University**, Mount Pleasant, MI, USA

*Aug 2024 - Dec 2025*

Master of Science in Computer Science, GPA: 3.87/4.0

**Lovely Professional University**, Punjab, India

*Aug 2016 - May 2020*

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

## **Certifications**

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Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate